Aymen Al-Tahmschi

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Education

University of Guelph **Expected Graduation: May 2028**
Bachelor of Engineering, Engineering Systems & Computing (Co-op) Guelph, ON

- **Relevant Coursework**: Computer Organization & Design, Digital Systems Design, Microcomputer Interfacing, Real-Time Systems Design, Embedded Reconfigurable Computing Systems, VLSI Digital Design

Selected Projects

16-bit CPU Design & Restoration | *VHDL, NVC* **2025; restored 2026**

- Restored a 16-bit VHDL CPU supporting nine decoded operations by integrating recovered ALU/control modules with a register file, program counter, and memory interface.
- Validated ALU behaviour with 278,544 result checks covering operand combinations, arithmetic boundaries, and full-width pseudorandom inputs.
- Passed three self-checking simulation benches, including a recovered 15-instruction program, reset behaviour, signed-comparison overflow, and program-counter wrap.

Hierarchical Six-bit Adder | *VHDL, GHDL* **Sep. 2026**

- Verified a hierarchical six-bit VHDL adder across all 8,192 operand and carry-in combinations using a self-checking testbench.
- Automated GHDL simulation through GitHub Actions and documented reconstructed RTL, board constraints, and repeatable verification steps.

ResolveAI | *TypeScript, React, Node.js, SQLite* **Sep. 2026**

- Built a full-stack troubleshooting application with 12 read-only macOS checks and 17 guided procedures, turning device evidence into structured investigations.
- Prevented stale incident updates through owner-scoped storage, input validation, and revision checks.
- Validated diagnostic and controlled-repair behaviour with 96 automated tests; documented reproducible setup and independent database write/read checks.

Work Experience

IT Service Desk Support Co-op **Jan. 2026 – Aug. 2026**
City of Guelph Guelph, ON

- Diagnosed hardware, software, account-access, and network issues for 100+ users weekly through Cherwell, remote sessions, telephone, and walk-in support.
- Documented troubleshooting steps, resolution details, and escalation context so other IT teams could continue unresolved cases without repeating completed work.
- Configured Windows 11 and Windows 10 laptops, printers, and peripherals; isolated device and connectivity faults across multiple city offices.
- Managed Active Directory accounts, password resets, mailboxes, distribution lists, licences, MFA requests, and employee offboarding tasks.

Machine Learning Code Engineer **Mar. 2025 – Present**
Outlier AI Remote

- Reviewed 250+ AI-generated code samples for correctness, efficiency, readability, and adherence to coding requirements.
- Identified flaws and optimized logic in 100+ implementations across Python, Java, C++, JavaScript, Swift, and Verilog.
- Created explanations, ranked model responses, and tagged evaluation data across 20+ AI training and assessment projects.

Extra Curriculars

Middle Eastern Students Association (MESA) **Feb. 2024 – Present**
Events Lead Guelph, ON

- Organized cultural events and coordinated sponsorship outreach, vendors, volunteers, and event logistics.

Guelph Cyber Security Society **Sep. 2023 – Present**
Technical Lead Guelph, ON

- Designed and led student workshops on penetration testing, ethical hacking, and threat detection.

Technical Skills

General: Leadership, Teamwork, Communication, Organization, Time Management, Problem Solving
Software: VHDL, C, C++, Python, Java, MATLAB, Assembly, TypeScript, SQL, Bash
Tools: NVC, GHDL, GitHub Actions, Git, FPGA Boards, Active Directory, Cherwell, MS Office & Excel, Linux, Windows 11/10